

VALENTINE, NE, USA

WMO: 725670

Lat: 42.857N

Lon: 100.548W

Elev: 2590

StdP: 13.37

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 24032

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.2	-3.3	-15.9	2.5	-8.5	-10.1	3.5	-2.2	31.9	24.6	27.8	26.9	7.7	280	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	97.9	69.3	94.1	68.9	90.6	68.4	74.0	88.0	72.4	86.6	70.9	85.0	14.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.0	121.6	79.9	68.2	114.0	78.8	66.4	107.2	77.6	39.2	88.1	37.7	86.5	36.3	84.9	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.9	24.9	22.0	DB	-20.0	103.7	5.8	3.7	-24.2	106.3	-27.6	108.5	-30.8	110.5	-35.1	113.2	(4)
(5)			WB	-20.1	76.6	5.5	1.4	-24.1	77.6	-27.3	78.4	-30.3	79.2	-34.3	80.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	24.2	26.7	36.8	46.5	57.3	68.2	75.4	73.1	63.7	49.0	36.1	25.4
	DBStd	20.82	12.84	13.12	12.71	10.46	8.97	7.57	6.56	6.25	9.25	9.88	11.33	11.97
	HDD50	3439	801	655	431	181	32	1	0	0	8	138	428	764
	HDD65	6836	1266	1073	874	556	269	51	5	11	133	503	866	1229
	CDD50	2949	0	2	22	78	260	548	786	715	420	106	12	0
	CDD65	871	0	0	0	3	31	148	327	262	95	5	0	0
	CDH74	11235	0	0	30	130	545	1862	4055	3013	1398	187	15	0
	CDH80	5222	0	0	3	35	185	778	2104	1441	619	55	2	0
(14) Wind	WSAvg	10.3	9.8	10.1	10.8	11.8	11.3	10.3	9.5	9.6	10.1	10.4	10.0	9.7
(15) Precipitation	PrecAvg	18.7	0.3	0.5	1.0	1.9	3.1	3.1	3.0	2.2	1.5	1.0	0.6	0.3
	PrecMax	32.7	0.8	1.4	4.2	5.5	9.0	7.1	9.0	6.7	5.9	3.5	2.6	1.8
	PrecMin	10.6	0.0	0.1	0.0	0.2	0.4	0.4	0.3	0.1	0.3	0.0	0.0	0.0
	PrecStd	4.8	0.2	0.4	0.8	1.2	1.7	1.6	1.7	1.3	1.1	0.8	0.6	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.5	67.8	78.7	85.0	91.5	98.6	104.1	100.3	97.4	86.6	75.8	60.9
		MCWB	44.2	47.6	52.6	56.5	64.8	68.4	69.2	69.1	65.5	59.5	51.8	45.0
	2%	DB	52.8	58.3	70.5	77.3	85.1	92.2	99.3	96.0	92.2	79.1	67.1	53.1
		MCWB	41.0	43.6	50.5	55.2	62.7	69.1	69.5	69.0	65.9	57.0	48.7	41.2
	5%	DB	46.9	51.2	63.8	71.6	80.1	88.2	95.0	92.1	87.3	73.2	60.5	47.2
		MCWB	37.8	39.9	48.2	53.0	61.1	67.7	70.1	68.8	64.6	55.4	45.8	38.0
	10%	DB	41.9	45.3	57.5	65.9	75.3	84.2	91.3	88.1	82.5	67.4	54.1	41.5
		MCWB	35.2	37.6	44.6	50.3	58.7	66.2	69.8	68.1	63.0	52.7	43.1	34.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.1	48.6	56.1	59.8	68.2	74.5	76.4	75.1	71.6	63.5	53.4	46.0
		MCDB	59.0	64.7	69.4	76.5	83.8	88.0	90.5	90.1	85.5	76.1	70.4	59.4
	2%	WB	41.3	44.5	51.9	56.6	65.5	71.7	74.7	73.0	69.0	59.8	49.8	41.6
		MCDB	52.1	56.6	67.8	73.9	80.3	85.7	88.8	87.0	84.3	74.4	65.1	51.9
	5%	WB	38.3	41.0	48.7	53.9	62.8	69.8	73.1	71.4	67.1	56.7	46.5	38.3
		MCDB	46.1	50.9	62.8	69.4	77.0	84.0	87.5	85.1	81.8	70.0	60.1	46.6
	10%	WB	35.4	37.6	45.3	51.3	60.1	68.0	71.6	70.0	64.9	53.5	43.2	35.1
		MCDB	41.2	45.2	57.3	63.7	72.5	81.4	86.4	83.2	78.5	65.8	53.5	41.2
(35) Mean Daily Temperature Range	5% DB	MDBR	23.7	23.6	25.2	24.9	24.4	25.2	26.4	26.6	28.0	26.7	26.1	23.3
		MCDBR	31.0	32.7	37.4	36.3	32.8	30.8	32.2	32.4	34.5	37.8	36.8	31.2
		MCWBR	20.3	19.7	20.2	17.9	14.7	12.3	10.6	11.3	13.3	18.3	20.3	20.4
	5% WB	MCDBR	29.2	31.2	34.4	32.5	28.1	26.6	27.3	27.1	28.9	32.7	35.0	29.9
		MCWBR	19.4	19.3	19.1	16.7	13.4	12.0	11.5	10.9	12.7	17.0	19.8	20.0
(40) Clear-Sky Solar Irradiance	taub		0.259	0.268	0.291	0.328	0.350	0.362	0.383	0.381	0.341	0.307	0.271	0.261
	taud		2.534	2.514	2.494	2.393	2.417	2.413	2.391	2.374	2.456	2.521	2.543	2.512
	Ebn at Noon		287	301	305	298	292	288	280	278	282	281	279	275
	Edh at Noon		21	26	30	36	36	36	37	36	31	25	21	20
(44) All-Sky Solar Radiation	RadAvg		662	942	1284	1605	1912	2175	2220	1882	1522	1037	718	552
	RadStd		46	59	104	97	159	144	77	104	92	108	52	41

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page